Roma

Can we test the method against real people's choices that it never saw?

Roma: *"Can we have a held-out set from Community Alignment and report performance of the method? We were struggling to figure out if there was a ground-truth preference test we could do."*
Answer. Yes. Community Alignment is a public dataset in which people pick their favorite of four responses to a prompt. In 81 of its sets of four, about 30 people each picked a favorite, so we know what share of people prefers each response. We held these sets out and asked each method to predict those shares. GPM-Pop, our model of the whole population's preferences, names the response most people picked in 56% of sets, where chance is 25%. Per-Fit, which fits a separate model to each annotator and averages them, does so in 40%.

Metric	Value	What it means
Picks the judges' favorite · GPM-Pop	**0.556**	Share of the 81 held-out sets where GPM-Pop's most likely response is the one most judges chose. Chance is 0.25.
Picks the judges' favorite · per-annotator models	**0.395**	The same measure for Per-Fit. Always picking the display slot people choose most often gives 0.333.
Cross-entropy against observed choice shares	**-0.047 [-0.077, -0.016]**	GPM-Pop minus Per-Fit; lower is better. Guessing uniformly over the four scores 1.386.

How we tested it. Each method is scored on the share of judges choosing each of the four responses, so no conversion to pairs is involved. People favor some display slots over others. GPM-Pop still improves on a model that knows only the slot (−0.028 [−0.046, −0.010] cross-entropy).

Daniel

Can we build a version of InTransit from Community Alignment, with a more accurate generator that can be checked against real people?

Daniel: *"Do some version of InTransit using the Community Alignment dataset ... InTransit is currently using a pretty simple method with relatively low accuracy, which makes it harder to validate."*
Answer. Yes, we built InTransit-CA. InTransit is our synthetic dataset: every simulated person

ranks every response, so the true share of people preferring one response over another is known. Its rankings come from a simple model (predicted attribute scores times each person's recorded value weights), and no real person's choices can confirm them. For InTransit-CA we fine-tuned a reward model, GPM-Gemma-2B, on 232,669 real favorite-of-four choices made by 3,603 Community Alignment annotators. Each simulated annotator then ranks responses by the preferences the model learned for that annotator. The fine-tuned model predicts annotators' choices on conversations it never saw better than the same model without fine-tuning. It also matches 481 held-out sets, each judged by about 31 real people, better than the model without fine-tuning.

Metric	Value	What it means
Log loss on held-out choices	**1.288 → 1.260**	On each annotator's later conversations: model without fine-tuning → fine-tuned. Difference −0.028 [−0.033, −0.023]; lower is better.
Top-1 accuracy on held-out choices	**0.396 → 0.431**	On English conversations: share of later choices where the model's top response is the one the annotator picked. +0.035 [+0.024, +0.046]; chance is 0.25.
Log loss on real judges' choices	**−0.019 [−0.026, −0.013]**	Fine-tuned minus not fine-tuned, on 481 held-out sets judged by about 31 people each, in five languages. Lower is better.

How we tested it. Where a response was displayed is modeled separately, so the model learns from content. InTransit-CA is released with the exact share of the population preferring each response in every pair, for 185,450 sets of four, and a data card.

With a fixed budget of votes, should we collect complete comparisons on fewer prompts or random pairs on more prompts?

Daniel: *"Given a fixed budget of data, would we rather have complete graphs over fewer prompts, or randomly sampled pairs across more prompts?" With the framing he asked for elsewhere: "it is impossible to validate any of these claims about accuracy, majority consistency, calibration because the datasets are much too sparse."*

Answer. To measure how well a reward model matches a population, spend the budget on complete comparisons with many votes per pair. Public datasets record one vote per pair of responses. With that format, a reward model's majority accuracy (how often it prefers the response most people prefer) comes out too low by 0.047 on InTransit and by 0.251 on Community Alignment. The 95% interval misses the true value every time, even at 100,000 votes. With 45 votes per pair, the error is 0.003 and the interval contains the true value 97% of the time. Cycles and Condorcet winners show up only when every pair in a set is compared

many times.

Metric	Value	What it means
Interval coverage, one vote per pair	**0%**	Share of 95% intervals that contain a reward model's true majority accuracy, at every budget up to 100,000 votes. With 45 votes per pair: 97%.
Error in majority accuracy, one vote per pair	**−0.047 / −0.251**	Estimated minus true majority accuracy at 100,000 votes, on InTransit / Community Alignment. With 45 votes per pair on InTransit: −0.003.
Share of InTransit sets with a cycle	**17% (truth 14%)**	What complete comparisons with 45 votes per pair report. With 5 votes per pair they report 56%; one vote per pair cannot see cycles.

How we tested it. We spent the same budget four ways: one vote per random pair, 45 votes per random pair, and every pair in fewer sets with 5 or with 45 votes. We did this on two populations where every true share is known, InTransit and one built from Community Alignment, and compared each estimate with the truth. This puts numbers on the paper's claim in §2 that sparse data cannot validate accuracy, calibration or cycles.

Does filling in missing preferences help predict real people, not just held-out synthetic data? And what is the actual Bradley-Terry loss?

Daniel: *"For InTransit ... it is asking if using more synthetic data is helpful for predicting held-out synthetic ... very different from Habermas, to see if our synthetic procedure can help predict true ground-truth human labels. That is very interesting if this truly happened. At minimum we should be reporting the true Bradley-Terry loss."*

Answer. Yes, on real people. In the Habermas dataset, panels of real jurors each rank several candidate statements on a political question. We trained a Bradley-Terry (BT) reward model, which gives each statement one score, in two ways. The first used one sparse comparison per panel. The second used data where our completion step filled in the missing comparisons. We graded both against the real jurors of 1,184 held-out panels. The model trained on completed data agrees with the majority more often (0.659 → 0.687) and has lower BT log loss (0.673 → 0.643). The gain holds on the 657 panels whose question the completion step never saw (0.674 → 0.705), so overlap between training and test questions doesn't explain it.

Metric	Value	What it means
Majority accuracy, all held-out panels	**0.659 → 0.687**	BT trained on one sparse label per panel → trained on completed data, graded against the real jurors of

		1,184 held-out panels. Mean of 3 seeds.
Majority accuracy, unseen questions only	**0.674 → 0.705**	The same comparison on the 657 panels whose question the completion step never saw.
BT log loss against real juror fractions	**0.673 → 0.643**	Sparse → completed, on held-out panels; lower is better. Predicting the true fractions scores 0.456.

How we tested it. In the paper this is now its own table. The InTransit table asks only whether more synthetic data helps predict held-out synthetic data.

Why don't we report log loss, the quantity GPM is trained to minimize?

Daniel: *"Why not report log loss? Isn't that what GPM is trying to fit to?"*
Answer. We now do, and GPM-Pop has the lowest log loss on both real datasets. Log loss scores how well predicted preference shares match the observed ones. Lower is better, and predicting 50/50 everywhere scores 0.693. Raw log loss heavily punishes a few overconfident predictions. So every method's predictions are pulled toward 50/50 by a factor fitted on other pairs, never on the pair being scored. After that, GPM-Pop scores 0.657 on Habermas, against 0.691 for Per-Fit (one model per annotator, averaged). On Community Alignment it scores 0.645 against 0.677. On InTransit it is within 0.008 of the two best methods.

Metric	Value	What it means
Log loss on held-out pairs, Habermas	**0.657 vs 0.691**	GPM-Pop vs Per-Fit, after the shared shrinkage; lower is better. Predicting 0.5 everywhere scores 0.693.
Log loss on held-out pairs, Community Alignment	**0.645 vs 0.677**	GPM-Pop vs Per-Fit; lower is better. Predicting 0.5 everywhere scores 0.693.
Log loss on held-out pairs, InTransit	**0.601**	GPM-Pop after the shared shrinkage, within 0.008 of PAL (0.593) and VPL (0.596). Predicting the true fractions scores 0.256.

How we tested it. Log loss is now a column of the completion table, both raw and after the shared shrinkage.

InTransit's responses were chosen to produce cycles. Do reward

models also struggle on cycles in real data we didn't design?

Daniel: *"We specifically chose responses in order to make cycles, so many of these numbers depend on design choices." (on the abstract's cycle claim)*
Answer. Yes. A cycle is three responses where a majority prefers A to B, B to C, and C to A. The Habermas dataset has 417 such cycles in real jurors' rankings of statements we didn't write. Cycles are close votes, and close votes are hard for any model. So we compared each cycle with the ten non-cyclic triples whose three votes are just as close. All four released reward models agree with the majority less often on the cycles, by 0.04 to 0.10.

Metric	Value	What it means
Extra loss on cycles, Habermas	**+0.04 to +0.10**	Agreement with the majority on equally close non-cyclic triples minus on cycles, for each of the four released reward models. Every interval excludes zero.
Largest gap · FsfairX-LLaMA3 (8B)	**+0.098 [+0.079, +0.117]**	Agreement 0.593 on the matched non-cyclic triples against 0.496 on cycles.
Cycles tested	**417**	Real Habermas majority cycles, each matched to the ten non-cyclic triples whose three votes are nearest in how far they are from 50/50.

Can the Rank-4 head predict cycles, unlike scalar reward models?

Daniel: *"So this can produce cycles, right? It doesn't have the same 0.67 cap as the others." (on the Rank-4 head)*
Answer. Yes. A scalar reward model gives each response one score, so on a cycle it can agree with at most two of the three majorities. That is the 0.67 cap. The Rank-4 head scores each response along four dimensions, so it can agree with all three. On InTransit it predicts about 165 cycles across 300 held-out sets. It is also the only model that agrees with the majority more on cycles than on non-cyclic triples that are just as close (0.542 vs 0.484). Scalar BT and IPO score 0.50–0.52 on cycles.

Metric	Value	What it means
Agreement on cycle edges, InTransit	**0.542 vs 0.484**	Rank-4 head on cycles vs on equally close non-cyclic triples. Difference 0.057 [0.022, 0.090]. Scalar BT and IPO score 0.50–0.52 on cycles.

Cycles it predicts	**≈165**	Triples in the 300 held-out sets whose predicted majorities form a cycle. A scalar head can predict none.
Majority accuracy / ECE	**0.700 / 0.127**	Rank-4 head trained on completed data, on held-out pairs; this cell was blank in Table 3. ECE is ten-bin calibration error; lower is better.

How many of the triples GPM-Pop flags as cycles are real cycles?

Daniel: *"We only report false negatives for cycles; should maybe also report false positives."*
Answer. On InTransit, 4.6% of flagged triples are real cycles, against 1.4% of all triples. A flag therefore makes a real cycle about three times as likely. GPM-Pop flags the 20% of triples whose closest predicted vote is nearest to 50/50. These flags catch 17 of the 22 real cycles, more than any other completion method (PAL 9, VPL 14). Without any threshold, GPM-Pop's predicted majorities form 5 cycles, and 2 of those are real.

Metric	**Value**	**What it means**
Precision of cycle flags, InTransit	**4.6% vs 1.4%**	Share of GPM-Pop's flagged triples that are real cycles, against the share of all held-out triples that are.
Recall of cycle flags	**17 / 22**	Real cycles among the flagged triples of 28 held-out sets; the most of five completion methods (PAL 9, VPL 14).
Cycles in its predicted majorities	**5, of which 2 real**	Triples whose predicted majorities themselves form a cycle, with no threshold. The other methods predict 0–1.

How we tested it. Precision and the number of predicted cycles are now columns of the completion table, next to recall, with one rule applied on every dataset.

Does the maximal lottery use the true (oracle) preference shares?

Daniel: *"Is this given the oracle pairwise proportions? This is a bit surprising to me." (on the maximal lottery)*
Answer. No. It uses the shares GPM-Pop predicts: predicted attribute scores times the annotators' recorded value weights. These agree with the true majority on 77.4% of held-out pairs. The maximal lottery is a voting rule. It picks one candidate response at random, with odds set so that, on average, no other candidate is preferred to its pick by a majority. Choosing

among 8 candidates, the lottery's pick sits higher in annotators' rankings than any reward model's pick (regret 0.219 vs 0.279). A majority prefers some other candidate in 54% of draws, against 68–69% for the reward models.

Metric	**Value**	**What it means**
Best-of-8 regret	**0.219 vs 0.279**	Maximal lottery vs the best reward model. Regret is how far down an annotator's ranking the pick falls, from 0 (their favorite) to 1 (their least favorite); lower is better.
Majority-rejected picks, best-of-8	**54.0% vs 68-69%**	Share of draws where a majority prefers another candidate: lottery vs every reward model. Lower is better.
Accuracy of the shares it reads	**0.774**	Majority accuracy of GPM-Pop's predicted shares on held-out pairs.